HUMAN SPACEFLIGHT: GAGANYAAN AND PLANETARY MISSIONS

HUMAN-RATING PYRAMID → GAGANYAAN MODULE MAP → CREW-SURVIVAL CHAIN → QUALIFICATION LADDER → INTERNATIONAL-EXPOSURE FIREWALL → MISSION-STATUS LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 HUMAN-RATING PYRAMID

HUMAN-RATING PYRAMID | LAUNCH RELIABILITY | NECESSARY STARTING CONDITION | MARGINS AND REDUNDANCY | TOLERATE SELECTED FAILURES | HEALTH MONITORING | DETECT UNSAFE VEHICLE CONDITIONS | ABORT MODES

LAUNCH RELIABILITY → necessary starting condition

MARGINS AND REDUNDANCY → tolerate selected failures → HEALTH MONITORING → detect unsafe vehicle conditions → ABORT MODES → protect crew during ascent

CONCEPT / ARCHITECTURE

- LAUNCH RELIABILITY → necessary starting condition
- MARGINS AND REDUNDANCY → tolerate selected failures

MECHANISM / APPLICATION

- HEALTH MONITORING → detect unsafe vehicle conditions
- ABORT MODES → protect crew during ascent

READINESS / STATUS LIMIT

- HUMAN RATING → system qualification, not a success count
- LAUNCH RELIABILITY → necessary starting condition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: HUMAN RATING → system qualification, not a success count.

01

STAGE 01 GAGANYAAN MODULE MAP

GAGANYAAN MODULE MAP | CREW MODULE | HABITABLE AND RE-ENTRY CAPABLE | SERVICE MODULE | PROPULSION, POWER AND SUPPORT | ORBITAL MODULE | CREW MODULE PLUS SERVICE MODULE | LAUNCH VEHICLE

CREW MODULE → habitable and re-entry capable

SERVICE MODULE → propulsion, power and support → ORBITAL MODULE → Crew Module plus Service Module → LAUNCH VEHICLE → ascent platform, separate from spacecraft

CONCEPT / ARCHITECTURE

- CREW MODULE → habitable and re-entry capable
- SERVICE MODULE → propulsion, power and support

MECHANISM / APPLICATION

- ORBITAL MODULE → Crew Module plus Service Module
- LAUNCH VEHICLE → ascent platform, separate from spacecraft

READINESS / STATUS LIMIT

- RULE → module, spacecraft and launcher are different layers
- CREW MODULE → habitable and re-entry capable

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → module, spacecraft and launcher are different layers.

02

STAGE 02 CREW-SURVIVAL CHAIN

CREW-SURVIVAL CHAIN | BREATHABLE AND THERMALLY CONTROLLED CABIN | CREW ESCAPE | SEPARATION FROM A FAILING LAUNCHER | CONTROLLED ATMOSPHERIC RETURN | STAGED DECELERATION | SPLASHDOWN, EXTRACTION AND MEDICAL SUPPORT

ECLSS → breathable and thermally controlled cabin → CREW ESCAPE → separation from a failing launcher → RE-ENTRY → controlled atmospheric return → PARACHUTES → staged deceleration → RECOVERY → splashdown, extraction and medical support

STARTING CONDITION / INPUT

- ECLSS → breathable and thermally controlled cabin
- CREW ESCAPE → separation from a failing launcher

SCIENTIFIC OR ENGINEERING MECHANISM

- RE-ENTRY → controlled atmospheric return
- PARACHUTES → staged deceleration

OUTPUT / FAILURE BOUNDARY

- RECOVERY → splashdown, extraction and medical support
- ECLSS → breathable and thermally controlled cabin

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CREW ESCAPE → separation from a failing launcher.

03

STAGE 03 QUALIFICATION LADDER

QUALIFICATION LADDER | GROUND TEST | VERIFY SUBSYSTEM BEHAVIOUR | AIR-DROP TEST | PARACHUTE AND RECOVERY SEQUENCE | TEST-VEHICLE ABORT | ESCAPE UNDER A SELECTED FLIGHT CONDITION | UNCREWED ORBITAL FLIGHT

COMPONENT / GROUND TEST → verify subsystem behaviour → AIR-DROP TEST → parachute and recovery sequence → TEST-VEHICLE ABORT → escape under a selected flight condition → UNCREWED ORBITAL FLIGHT → integrated mission without crew → CREWED FLIGHT → later accepted human risk

STARTING CONDITION / INPUT

SCIENTIFIC OR ENGINEERING MECHANISM

OUTPUT / FAILURE BOUNDARY

STARTING CONDITION / INPUT
- ECLSS → breathable and thermally controlled cabin
- CREW ESCAPE → separation from a failing launcher

SCIENTIFIC OR ENGINEERING MECHANISM
- RE-ENTRY → controlled atmospheric return
- PARACHUTES → staged deceleration

OUTPUT / FAILURE BOUNDARY
- RECOVERY → splashdown, extraction and medical support
- ECLSS → breathable and thermally controlled cabin

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CREW ESCAPE → separation from a failing launcher.

03

STAGE 03

QUALIFICATION LADDER

QUALIFICATION LADDER · GROUND TEST · VERIFY SUBSYSTEM BEHAVIOUR · AIR-DROP TEST · PARACHUTE AND RECOVERY SEQUENCE · TEST-VEHICLE ABORT · ESCAPE UNDER A SELECTED FLIGHT CONDITION · UNCREWED ORBITAL FLIGHT

COMPONENT / GROUND TEST → verify subsystem behaviour → AIR-DROP TEST → parachute and recovery sequence → TEST-VEHICLE ABORT → escape under a selected flight condition → UNCREWED ORBITAL FLIGHT → integrated mission without crew → CREWED FLIGHT → later accepted human risk

STARTING CONDITION / INPUT
- COMPONENT / GROUND TEST → verify subsystem behaviour
- AIR-DROP TEST → parachute and recovery sequence

SCIENTIFIC OR ENGINEERING MECHANISM
- TEST-VEHICLE ABORT → escape under a selected flight condition
- UNCREWED ORBITAL FLIGHT → integrated mission without crew

OUTPUT / FAILURE BOUNDARY
- CREWED FLIGHT → later accepted human risk
- COMPONENT / GROUND TEST → verify subsystem behaviour

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UNCREWED ORBITAL FLIGHT → integrated mission without crew.

04

STAGE 04

INTERNATIONAL-EXPOSURE FIREWALL

INTERNATIONAL-EXPOSURE FIREWALL · INTERNATIONAL ISS MISSION EXPERIENCE · FOREIGN LAUNCHER · EXTERNAL TRANSPORT SYSTEMS · INDIAN EXPERIMENTS · OPERATIONAL AND SCIENTIFIC EXPOSURE · INDIGENOUS LAUNCH, MODULE AND RECOVERY OBJECTIVE · EXPERIENCE COMPLEMENTS BUT DOES NOT CERTIFY CAPABILITY

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
AXIOM-4 → international ISS mission experience	INDIAN EXPERIMENTS → operational and scientific exposure	RULE → experience complements but does not certify capability
FOREIGN LAUNCHER / SPACECRAFT → external transport systems	GAGANYAAN → indigenous launch, module and recovery objective	AXIOM-4 → international ISS mission experience

SYSTEM / DEVICE / INSTITUTION
- AXIOM-4 → international ISS mission experience
- FOREIGN LAUNCHER / SPACECRAFT → external transport systems

DECISIVE TECHNICAL DISTINCTION
- INDIAN EXPERIMENTS → operational and scientific exposure
- GAGANYAAN → indigenous launch, module and recovery objective

STATUS / UNIT / EVIDENCE LIMIT
- RULE → experience complements but does not certify capability
- AXIOM-4 → international ISS mission experience

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → experience complements but does not certify capability.

05

STAGE 05

MISSION-STATUS LADDER

MISSION-STATUS LADDER · AUTHORITY AND PROGRAMME SANCTION · UNDER DEVELOPMENT · WORK AND EVIDENCE RUNG · DISTINCT FLIGHT MILESTONES · OUTCOME VOCABULARY

ANNOUNCED / PROPOSED → intent

APPROVED → authority and programme sanction → UNDER DEVELOPMENT / TESTED → work and evidence rung → LAUNCHED / INSERTED / OPERATIONAL → distinct flight milestones

STARTING CONDITION / INPUT
- ANNOUNCED / PROPOSED → intent
- APPROVED → authority and programme sanction

SCIENTIFIC OR ENGINEERING MECHANISM
- UNDER DEVELOPMENT / TESTED → work and evidence rung
- LAUNCHED / INSERTED / OPERATIONAL → distinct flight milestones

OUTPUT / FAILURE BOUNDARY
- COMPLETED / PARTIAL / FAILED / CONCLUDED → outcome vocabulary
- ANNOUNCED / PROPOSED → intent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COMPLETED / PARTIAL / FAILED / CONCLUDED → outcome vocabulary.

06

STAGE 06

LUNAR CAPABILITY LADDER

LUNAR CAPABILITY LADDER · ORBITAL SCIENCE AND LUNAR-WATER EVIDENCE · ORBITER SUCCESS, LANDER SETBACK · SOFT LANDING AND SURFACE OPERATIONS · APPROVED SAMPLE-RETURN TECHNOLOGY MISSION · LANDING DOES NOT PROVE ASCENT, DOCKING OR EARTH RETURN

CHANDRAYAAN-1 → orbital science and lunar-water evidence → CHANDRAYAAN-2 → orbiter success, lander setback → CHANDRAYAAN-3 → soft landing and surface operations → CHANDRAYAAN-4 → approved sample-return technology mission → RULE → landing does not prove ascent, docking or Earth return

STARTING CONDITION / INPUT
- CHANDRAYAAN-1 → orbital science and lunar-water evidence
- CHANDRAYAAN-2 → orbiter success, lander setback

SCIENTIFIC OR ENGINEERING MECHANISM
- CHANDRAYAAN-3 → soft landing and surface operations
- CHANDRAYAAN-4 → approved sample-return technology mission

OUTPUT / FAILURE BOUNDARY
- RULE → landing does not prove ascent, docking or Earth return
- CHANDRAYAAN-1 → orbital science and lunar-water evidence

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COMPLETED / PARTIAL / FAILED / CONCLUDED → outcome vocabulary.

06

STAGE 06

LUNAR CAPABILITY LADDER

LUNAR CAPABILITY LADDER | ORBITAL SCIENCE AND LUNAR-WATER EVIDENCE | ORBITER SUCCESS, LANDER SETBACK | SOFT LANDING AND SURFACE OPERATIONS | APPROVED SAMPLE-RETURN TECHNOLOGY MISSION | LANDING DOES NOT PROVE ASCENT, DOCKING OR EARTH RETURN

CHANDRAYAAN-1 → orbital science and lunar-water evidence → CHANDRAYAAN-2 → orbiter success, lander setback → CHANDRAYAAN-3 → soft landing and surface operations → CHANDRAYAAN-4 → approved sample-return technology mission → RULE → landing does not prove ascent, docking or Earth return

STARTING CONDITION / INPUT

- CHANDRAYAAN-1 → orbital science and lunar-water evidence
- CHANDRAYAAN-2 → orbiter success, lander setback

SCIENTIFIC OR ENGINEERING MECHANISM

- CHANDRAYAAN-3 → soft landing and surface operations
- CHANDRAYAAN-4 → approved sample-return technology mission

OUTPUT / FAILURE BOUNDARY

- RULE → landing does not prove ascent, docking or Earth return
- CHANDRAYAAN-1 → orbital science and lunar-water evidence

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → landing does not prove ascent, docking or Earth return.

07

STAGE 07

SOLAR-OBSERVATORY GEOMETRY

SOLAR-OBSERVATORY GEOMETRY | SUN-EARTH SYSTEM | DEFINES LAGRANGE-POINT GEOMETRY | L1 REGION | CONTINUOUS SOLAR-VIEW ADVANTAGE | HALO ORBIT | THREE-DIMENSIONAL PATH AROUND L1 | STATION KEEPING

SUN-EARTH SYSTEM → defines Lagrange-point geometry → L1 REGION → continuous solar-view advantage → HALO ORBIT → three-dimensional path around L1 → STATION KEEPING → periodic correction and propellant use → ADITYA-L1 → solar observatory, not a landing mission

STARTING CONDITION / INPUT

- SUN-EARTH SYSTEM → defines Lagrange-point geometry
- L1 REGION → continuous solar-view advantage

SCIENTIFIC OR ENGINEERING MECHANISM

- HALO ORBIT → three-dimensional path around L1
- STATION KEEPING → periodic correction and propellant use

OUTPUT / FAILURE BOUNDARY

- ADITYA-L1 → solar observatory, not a landing mission
- SUN-EARTH SYSTEM → defines Lagrange-point geometry

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ADITYA-L1 → solar observatory, not a landing mission.

08

STAGE 08

PLANETARY-STATUS MATRIX

PLANETARY-STATUS MATRIX | MARS ORBITER MISSION | HISTORIC MISSION, NOW CONCLUDED | VENUS ORBITER MISSION | APPROVED FUTURE MISSION IN OWNER | LUNAR SAMPLE RETURN | ASCENT AND RETURN ARCHITECTURE | SOLAR OBSERVATORY

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
MARS ORBITER MISSION → historic mission, now concluded	LUNAR SAMPLE RETURN → ascent and return architecture	RULE → destination and status both control the description
VENUS ORBITER MISSION → approved future mission in owner	SOLAR OBSERVATORY → remote sensing from L1 halo orbit	MARS ORBITER MISSION → historic mission, now concluded

SYSTEM / DEVICE / INSTITUTION

- MARS ORBITER MISSION → historic mission, now concluded
- VENUS ORBITER MISSION → approved future mission in owner

DECISIVE TECHNICAL DISTINCTION

- LUNAR SAMPLE RETURN → ascent and return architecture
- SOLAR OBSERVATORY → remote sensing from L1 halo orbit

STATUS / UNIT / EVIDENCE LIMIT

- RULE → destination and status both control the description
- MARS ORBITER MISSION → historic mission, now concluded

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → destination and status both control the description.

09

STAGE 09

PARALLEL CAPABILITY LADDERS

PARALLEL CAPABILITY LADDERS | ABORT, LIFE SUPPORT, UNCREWED, CREWED, STATION | ORBIT, SOFT LAND, ROVE, SAMPLE RETURN | TRANSIT, HALO INSERTION, STATION KEEPING, OBSERVATION | CRUISE, INSERTION, SCIENCE OPERATIONS | ONE LADDER CANNOT CERTIFY ANOTHER

HUMAN → abort, life support, uncrewed, crewed, station

LUNAR → orbit, soft land, rove, sample return → SOLAR → transit, halo insertion, station keeping, observation → PLANETARY → cruise, insertion, science operations

STARTING CONDITION / INPUT

- HUMAN → abort, life support, uncrewed, crewed, station
- LUNAR → orbit, soft land, rove, sample return

SCIENTIFIC OR ENGINEERING MECHANISM

- SOLAR → transit, halo insertion, station keeping, observation
- PLANETARY → cruise, insertion, science operations

OUTPUT / FAILURE BOUNDARY

- RULE → one ladder cannot certify another
- HUMAN → abort, life support, uncrewed, crewed, station

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOLAR → transit, halo insertion, station keeping, observation.

09 STAGE 09 PARALLEL CAPABILITY LADDERS

PARALLEL CAPABILITY LADDERS | ABORT, LIFE SUPPORT, UNCREWED, CREWED, STATION | ORBIT, SOFT LAND, ROVE, SAMPLE RETURN | TRANSIT, HALO INSERTION, STATION KEEPING, OBSERVATION | CRUISE, INSERTION, SCIENCE OPERATIONS | ONE LADDER CANNOT CERTIFY ANOTHER

HUMAN → abort, life support, uncrewed, crewed, station

LUNAR → orbit, soft land, rove, sample return → SOLAR → transit, halo insertion, station keeping, observation → PLANETARY → cruise, insertion, science operations

STARTING CONDITION / INPUT
- HUMAN → abort, life support, uncrewed, crewed, station
- LUNAR → orbit, soft land, rove, sample return

SCIENTIFIC OR ENGINEERING MECHANISM
- SOLAR → transit, halo insertion, station keeping, observation
- PLANETARY → cruise, insertion, science operations

OUTPUT / FAILURE BOUNDARY
- RULE → one ladder cannot certify another
- HUMAN → abort, life support, uncrewed, crewed, station

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SOLAR → transit, halo insertion, station keeping, observation.

10 STAGE 10 PLANETARY-DEFENCE RAIL

PLANETARY-DEFENCE RAIL | DISCOVER POTENTIALLY HAZARDOUS OBJECTS | REFINE ORBIT, SIZE AND COMPOSITION | WARNING TIME | DETERMINES RESPONSE FEASIBILITY | KINETIC IMPACT IS ONE DEMONSTRATED CONCEPT | RESIDUAL RISK | ONE TEST IS NOT A COMPLETE SHIELD

DETECTION → discover potentially hazardous objects

CHARACTERISATION → refine orbit, size and composition → WARNING TIME → determines response feasibility → DEFLECTION → kinetic impact is one demonstrated concept

STARTING CONDITION / INPUT
- DETECTION → discover potentially hazardous objects
- CHARACTERISATION → refine orbit, size and composition

SCIENTIFIC OR ENGINEERING MECHANISM
- WARNING TIME → determines response feasibility
- DEFLECTION → kinetic impact is one demonstrated concept

OUTPUT / FAILURE BOUNDARY
- RESIDUAL RISK → one test is not a complete shield
- DETECTION → discover potentially hazardous objects

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESIDUAL RISK → one test is not a complete shield.

11 STAGE 11 MISSION ANSWER SPINE

MISSION ANSWER SPINE | HUMAN LUNAR SOLAR PLANETARY OR DEFENCE MISSION | MINIMUM TECHNICAL AND SAFETY CHAIN | ONE VERIFIED ACHIEVEMENT OR APPROVED NEXT RUNG | TEST LAUNCH OPERATION COMPLETION AND OUTCOME | DATED CREW SCHEDULE PAYLOAD FINDING AND STATUS SOURCE

CLASSIFY → human lunar solar planetary or defence mission → TRACE → minimum technical and safety chain → NAME → one verified achievement or approved next rung → SEPARATE → test launch operation completion and outcome → VERIFY → dated crew schedule payload finding and status source

CONCEPT / ARCHITECTURE
- CLASSIFY → human lunar solar planetary or defence mission
- TRACE → minimum technical and safety chain

MECHANISM / APPLICATION
- NAME → one verified achievement or approved next rung
- SEPARATE → test launch operation completion and outcome

READINESS / STATUS LIMIT
- VERIFY → dated crew schedule payload finding and status source
- CLASSIFY → human lunar solar planetary or defence mission

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → dated crew schedule payload finding and status source.

E EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | CORE HUMAN-SPACEFLIGHT COORDINATING BODY FOR GAGANYAAN-RELATED ACTIVITY. | TRAINING AND RECOVERY NETWORK | ASTRONAUT TRAINING, SIMULATION, AERO-MEDICAL SUPPORT AND NAVAL RECOVERY ARE | MISSION-SCIENCE ECOSYSTEM | LUNAR, SOLAR AND PLANETARY MISSIONS DRAW ON TRACKING | PAYLOAD SCIENCE, PROPULSION AND MISSION-OPERATIONS CAPABILITIES ACROSS ISRO CENTRES. | HUMAN-SPACEFLIGHT PROGRAMMES ARE OFTEN CRITICISED AS PRESTIGE PROJECTS, BUT

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH
- HSFC/ISRO: core human-spaceflight coordinating body for Gaganyaan-related activity.
- Training and recovery network: astronaut training, simulation, aero-medical support and naval recovery are all institutional parts of the programme.
- Mission-science ecosystem: lunar, solar and planetary missions draw on tracking,
- payload science, propulsion and mission-operations capabilities across ISRO centres.

READINESS, UNIT OR STATUS LIMIT
- CAUTION: Human-spaceflight programmes are often criticised as prestige projects, but the
- deeper policy question is whether they generate durable systems capability.
- CAUTION: Space-station ambition raises governance questions of cost, research prioritisation, international collaboration and long-term operational sustainability.
- CAUTION: Planetary missions must be assessed not only by symbolism but by data return, technology maturation and downstream industrial learning.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE
- CAUTION: Mission sequencing matters: over-ambitious jumps can create schedule slippage, while overly conservative sequencing can reduce strategic momentum.
- Gaganyaan creates autonomous human-spaceflight capability rather than dependence on external launchers for crewed access.
- BAS planning signals a shift from one-off crewed demonstration to sustained orbital presence ambition.
- Chandrayaan progression shows India moving from scientific discovery to technology demonstration and sample-return aspiration.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels